Wind, slope, and the want of a tail: a computational feasibility study of Eilmer of Malmesbury's flight

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Abstract

Around 1000–1045 a monk at Malmesbury Abbey fixed wings to his hands and feet, glided from a tower for "more than a furlong" (about 200 m), crashed, broke both legs, and blamed the failure on having no tail. We test that account as a constrained engineering problem using terrain reconstruction, twenty years of ERA5 wind climatology, a first-order ridge-lift model, a glide-and-stability analysis, a 40,000-sample Monte Carlo over the unknown apparatus, and an impact-biomechanics model. The abbey sits on a SW-facing escarpment aligned with the prevailing wind, which is favourable for launch in 15.6 % of hours per year. A still-air glide would demand a sailplane-class glide ratio of 11.1, but ridge lift over the escarpment brings the requirement within reach of a crude cambered wing (L/D 3–7) above roughly 3.6 m s^{-1} of wind. A tailless hand-and-foot wing is statically unstable across every centre of gravity a hanging pilot can reach, and its pitch error doubles in a median 0.27 s against a flight lasting some 20 s; adding a tail removes the divergence. The predicted out-of-control arrival breaks both legs yet remains survivable, matching the reported injury. The account is internally consistent and Eilmer's own diagnosis is correct. We release the full pipeline so the method can adjudicate other pre-modern flight claims.

1 Introduction

In the eleventh century a monk of Malmesbury Abbey named Eilmer fixed wings to his hands and feet, leapt from a tower, glided for "more than a furlong," and crashed, breaking both legs and walking lame for the rest of his life. He later said the fault was his own: he had fitted no tail. This paper asks whether that account is physically self-consistent, and finds that it is. A wind-and-slope-assisted glide of about 200 m from the abbey is feasible for a crude cambered wing; a tailless hand-and-foot apparatus is statically unstable for any pilot posture and loses pitch control within a second of a disturbance; and the resulting out-of-control impact breaks both legs while remaining survivable. Eilmer's own diagnosis, the want of a tail, is the correct one.

The source is good by the standards of medieval history. William of Malmesbury, writing about a century later in the same house, knew men who had known Eilmer, and records concrete and testable details: a tower launch, wings on the hands and feet, a breeze, a glide of a furlong, a fall, broken legs, and the missing tail [1, 2]. Each of these is an engineering constraint. The question is not whether a medieval monk *could have* understood aerodynamics, but whether the event he is reported to have survived is consistent with the aerodynamics we understand now.

We treat the flight as a constrained engineering problem and attack it with the tools available in 2026: a digital terrain model of the abbey promontory, two decades of reanalysis wind

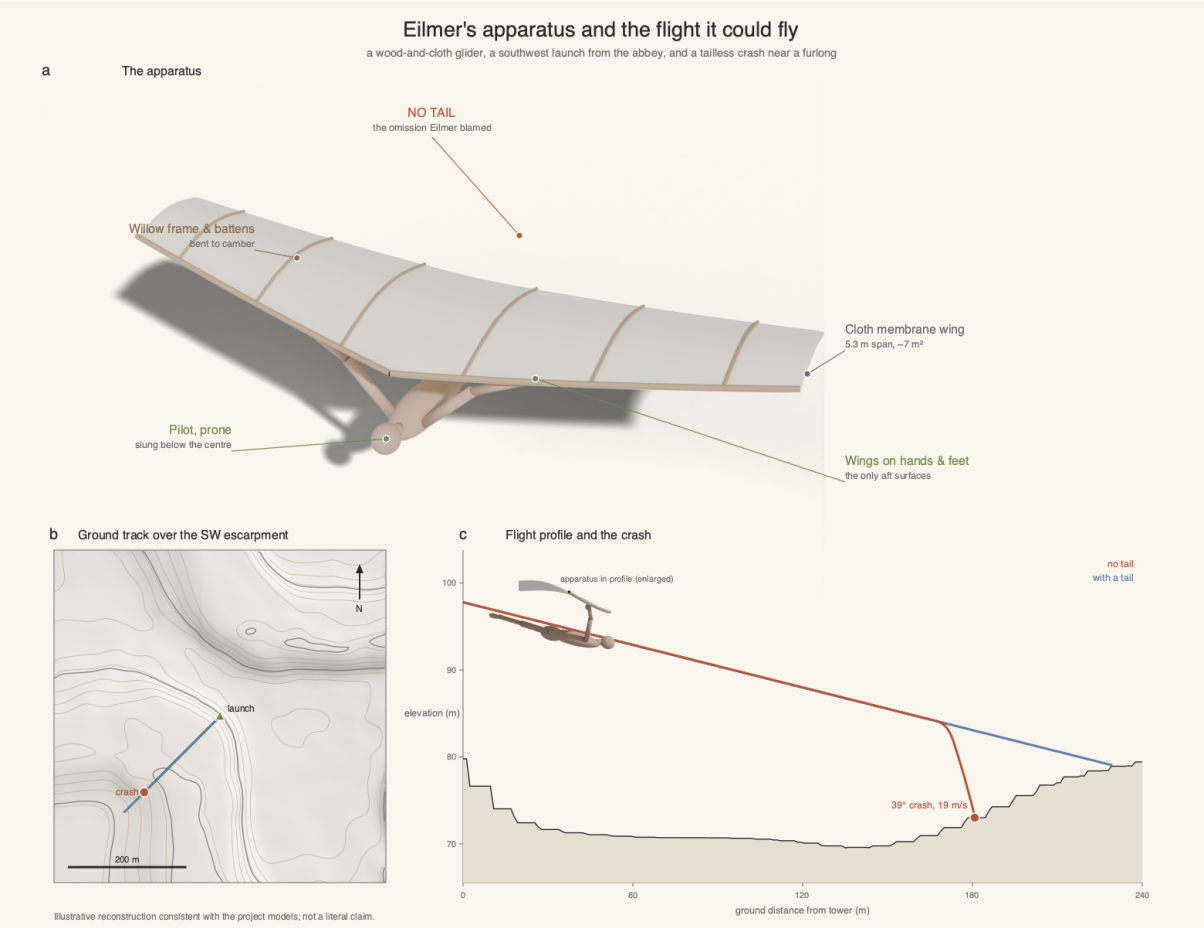


Figure 1: Eilmer's apparatus and the flight it could fly. **(a)** A to-scale reconstruction of the apparatus implied by the glide and stability models: a cambered cloth-membrane wing of about $7\,\mathrm{m}^2$ and 5.3 m span (aspect ratio 4) on a willow frame, the pilot prone with wings fixed to the hands and feet and, as William of Malmesbury records, no tail. **(b)** Launch geometry on the abbey promontory, with the southwest ground track of the glide over a contour map of the escarpment, from the tower to an impact about a furlong away. **(c)** The flight in profile, apparatus enlarged for clarity: a wind-and-slope-assisted glide with a tail (blue) against the tailless case (red), which loses pitch control and strikes the ground at about 39° and $19\,\mathrm{m\,s}^{-1}$. The reconstruction is illustrative and consistent with the project models, not a literal depiction.

climatology, a first-order orographic lift model, a glide and short-period pitch-stability analysis, a Monte Carlo over the unknown apparatus geometry and launch day, and an impact-biomechanics model calibrated to human long-bone fracture tolerances. Figure 1 summarises the reconstruction and the result. The contribution is twofold: a calibrated verdict on a specific historical claim, and a reproducible method for adjudicating pre-modern flight accounts from terrain, climate, and first-principles flight dynamics rather than from anecdote. Everything here rebuilds from raw data with a single command, and the apparatus we infer is a consequence of the models, not an input to them.

2 The source, the date, and prior work

William of Malmesbury reports the flight in his *Gesta Regum Anglorum* (about 1125), within living memory of the abbey's older monks and roughly a century after the event [1]. He names Eilmer, describes the apparatus as wings on the hands and feet, gives the distance as more than a stadium (a furlong, about 200 m), reports that wind or turbulence and his own fear brought him down, and that he broke both legs. He adds Eilmer's retrospective explanation that he had forgotten to fit a tail. This is the canonical account, and Lynn White's study remains the standard historical treatment [2]; later popular and survey histories follow it [3–5], and the *Oxford Dictionary of National Biography* dates Eilmer to about 985 and after 1066 [6].

The date of the flight is an open and actively debated question. The traditional reconstruction ties Eilmer's later comet sighting to the returns of Halley's comet and places the flight around 1000–1010 [7, 8]. Aitcheson [9], writing this year in *Notes and Queries*, reopens it: the earlier comet Eilmer recalled may have been the comet of 1018 rather than the 989 apparition of Halley, which would move his birth into the 990s to 1010s and the flight itself as late as the 1020s, 1030s, or early 1040s, decades later than the standard date. The window is therefore wider and less settled than the textbooks imply, and we do not try to close it. Because the flight depends on weather rather than on the calendar year, we treat the launch day as a variable sampled from the wind climatology of the site (Section 4). That choice makes our analysis insensitive to the dating dispute: the relevant prevailing-wind regime is stable across the whole eleventh-century window Aitcheson admits, so the feasibility verdict holds whichever decade the leap belongs to.

Human-flight attempts before the modern era are mostly legend, and the historiography is careful to separate reportable fact from myth-making [10]. Eilmer is unusual in having a credible near-contemporary witness and a named, physical cause of failure. The closest antecedent is the ninth-century Andalusian polymath Abbas Ibn Firnas, who is reported to have glided and likewise landed badly, traditionally also blamed on the lack of a tail [11]. That two of the earliest credible accounts converge on the same aerodynamic diagnosis is itself worth a quantitative test. The name of the modern open-source hypersonic flow solver *Eilmer* honours this same monk [12, 13], a small reminder that the aerodynamics community already claims him.

3 Methods

The analysis is split into independent workstreams, each writing a `metrics.json` that feeds both its figure and the numbers quoted here, so a single source produces every value. The whole pipeline rebuilds from raw data with one driver script.

3.1 Terrain

We build a digital elevation model of the abbey promontory from AWS Terrain Tiles (terrarium encoding, zoom 14, about 6 m posting). From it we compute the local slope and aspect within

a 400 m radius of the tower and extract an elevation section along the dominant SW–NE axis. The 6 m posting under-resolves the sharp launch edge, so slope estimates are reported as lower bounds pending airborne LIDAR.

3.2 Wind climatology

We take twenty years of ERA5 hourly 10 m wind at the abbey (2004–2023, 175 320 hours) through the Open-Meteo archive [14, 15]. A launch is counted favourable when the wind lies in the SW sector (191°–259°, aligned with the escarpment aspect) at 5–12 m s^{-1}. We report the favourable fraction annually and by meteorological season with 95 % confidence intervals taken across the twenty years. Reconstructions of the medieval North Atlantic Oscillation differ on timing, from a persistently positive Medieval Climate Anomaly mode [16] to one with positive phases concentrated in later centuries [17], but neither points to the weak westerlies of a negative phase, so modern reanalysis is a reasonable and probably conservative proxy for Eilmer's epoch; the Twentieth Century Reanalysis provides an independent stationarity check [18].

3.3 Ridge lift

We model orographic lift over the real terrain section with a first-order linear solution in which the vertical air velocity follows the surface slope and decays with height,

$$w(x, z) = U \, \frac{\mathrm{d}h}{\mathrm{d}x} \, e^{-kz}, \tag{1}$$

where U is the incoming wind speed, $\mathrm{d}h/\mathrm{d}x$ the along-wind surface slope, z height above the surface, and $k = 2\pi/L$ a decay rate set by the escarpment horizontal scale L. We compare the resulting lift against a crude glider's sink rate to estimate the depth of the soarable band. Equation (1) is a deliberately conservative stand-in for a microscale large-eddy or WAsP solve, which is deferred to a LIDAR-resolved follow-up.

3.4 Glide

Steady gliding flight sets the airspeed and sink rate. For mass m, wing area S, trim lift coefficient C_L, and air density ρ, the trim airspeed is $V = \sqrt{2mg/(\rho S C_L)}$, the glide angle γ satisfies $\tan\gamma = 1/(L/D)$, and the still-air sink is $V \sin\gamma$. With orographic upwash w the net sink is $V \sin\gamma - w$, and the ground range from launch height h is $R = V \cos\gamma \cdot h / \max(V \sin\gamma - w,\, 0)$, extended without limit where the net sink is non-positive (the pilot soars). We invert $R = 200$ m to find the glide ratio required as a function of wind speed, and compare it against the crude cambered-wing band (L/D 3–7) supported by low-Reynolds-number airfoil data [19–22].

3.5 Pitch stability

Longitudinal stability turns on the centre of gravity x_{cg} relative to the neutral point x_{np}, both in fractions of the mean chord. A hanging pilot's CG lies well aft, between 0.35 and 0.55 c. A tailless cambered wing has its neutral point near the quarter chord ($x_{\mathrm{np}} \approx 0.25\,c$); a tail shifts it aft and adds pitch damping. We linearise the short-period mode and form the state matrix in angle of attack and pitch rate, whose largest real eigenvalue λ sets the growth rate; the time for a pitch disturbance to double is $t_2 = \ln 2/\lambda$. We also integrate the nonlinear pitch response to an 8° gust and record the time to cross a 60° loss-of-control threshold, comparing it against the flight duration. The treatment follows standard flight-dynamics theory [21].

3.6 Monte Carlo synthesis

The apparatus geometry and the launch day are unknown, so we sample the plausible space with 40 000 draws (fixed seed). We draw wing area (5–10 m^2), all-up mass (70–90 kg), trim C_L (0.7–1.1), glide ratio (3–7), tower height (15–25 m), escarpment slope (8°–18°), ridge-lift efficiency (0.5–1.0), CG (0.35–0.55 c), aspect ratio (3–5), and the neutral-point shift from a tail (0.18–0.32 c). Launch wind is drawn from the actual ERA5 SW-sector launchable speeds rather than assumed, tying the synthesis to the measured climatology. Each world is scored on whether the glide covers 200 m and whether it stays controlled (t_2 longer than the flight) for the tailless and tailed configurations; we report Wilson confidence intervals on the fractions.

3.7 Impact biomechanics

We treat "broke both legs, survived, lame" as data. A feet-first impact decelerates the body over a stopping distance d made of leg flexion and ground give, so the average total force is energy over distance,

$$F = \frac{mv^2}{2d}, \qquad \text{per leg } F/2, \tag{2}$$

for vertical impact speed v. The tibia fractures under axial and combined loading at 7.5–11.3 kN per leg, which sets a fracture-onset speed $v^\star = \sqrt{2dF_{\mathrm{tot}}/m}$ [23–27]. We compare a flared touchdown (vertical speed near the sink rate) against an uncontrolled departure (vertical speed $V \sin\theta$ for impact angle θ) across the plausible range of ground give. Equation (2) is an average-force model, so the onset speeds are conservative upper bounds on what the legs can take; real peak transients and bending loads would lower the threshold.

4 Results

We present the empirical preconditions first (terrain, wind, lift), then the model-dependent flight dynamics and synthesis.

4.1 The abbey sits on a wind-aligned escarpment

The abbey occupies a promontory 10.2 m above the valley floor at the confluence of two branches of the Bristol Avon (Fig. 2). The local slope faces southwest (aspect 230°), with a mean gradient of 5.2° and a 90th-percentile gradient of 12.8° within 400 m of the tower. A prevailing southwest wind therefore strikes the slope close to perpendicular and is deflected upward, which is the geometric precondition for ridge lift. Because the 6 m DEM smooths the launch edge, these gradients are lower bounds.

4.2 Favourable launch winds are common

Over twenty years of ERA5 reanalysis, the wind blows from the SW sector at a launchable 5–12 $\mathrm{m\,s^{-1}}$ in 15.6 % of all hours (95 % CI 14.3–17.0), about 1370 hours per year, with a winter peak of 21.1 % (Fig. 3). The SW sector at any speed accounts for 34.3 % of hours, so the prevailing wind and the escarpment are well aligned. A would-be flyer who waits for his conditions has them on the order of a thousand hours a year, and the medieval wind regime appears no weaker than today's, so this estimate is conservative for Eilmer's time.

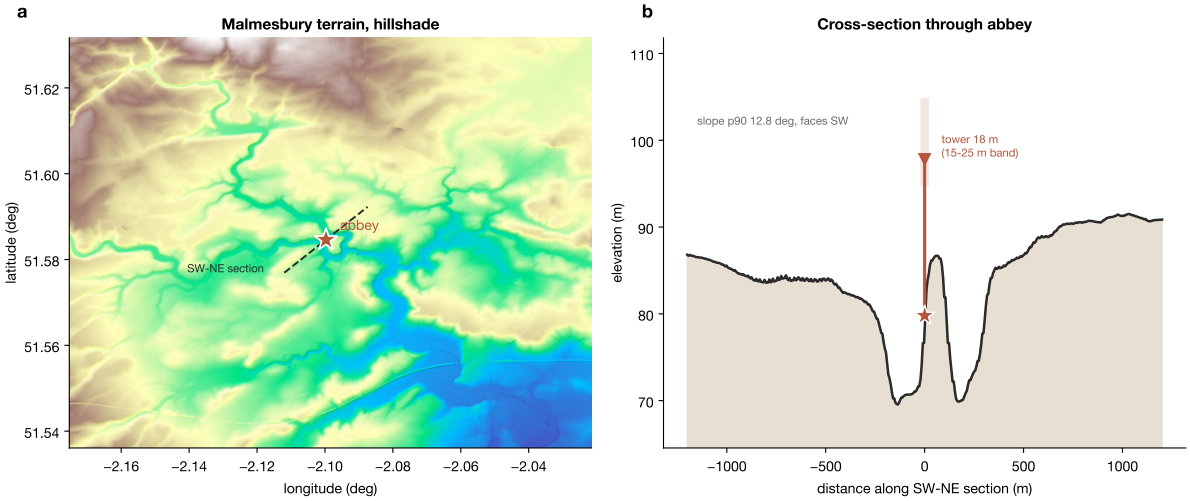


Figure 2: Site and launch geometry at Malmesbury Abbey. (a) Hillshaded terrain (AWS Terrain Tiles, about 6 m posting) with the abbey on a promontory above the confluence of two Avon branches; the SW–NE line marks the section in (b). **(b)** Elevation along the section. The abbey stands 10.2 m above the valley floor on a SW-facing slope (aspect 230°; slope within 400 m: mean 5.2°, p90 12.8°). The 6 m DEM under-resolves the launch edge, so the reported slope is a lower bound pending LIDAR.

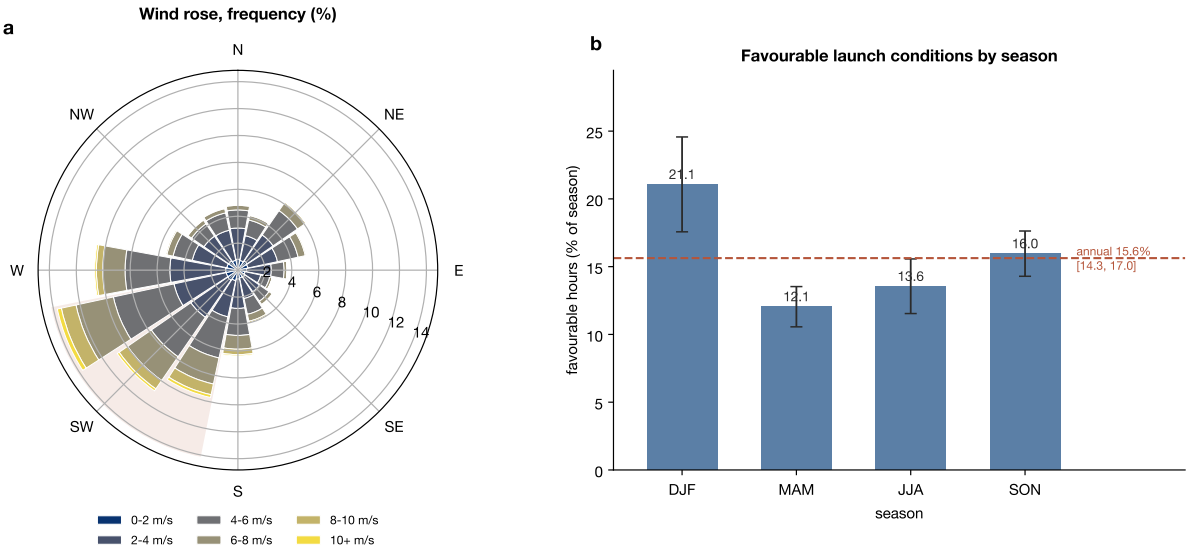


Figure 3: Favourable launch winds are common. (a) Wind rose of ERA5 hourly 10 m wind at the abbey, 2004–2023 (175 320 hours). **(b)** Fraction of hours offering a favourable launch (SW sector 191°–259° at 5–12 m s^{-1}), annually and by season, with 95 % confidence intervals across the twenty years. Favourable conditions occur in 15.6 % of hours (95 % CI 14.3–17.0), about 1370 hours per year, peaking in winter (21.1 %).

4.3 Ridge lift supplies the missing glide performance

The linear orographic model (Eq. (1)) over the real terrain section gives a peak upwash of about $1.5\,\mathrm{m\,s^{-1}}$ just southwest of the abbey for an $8\,\mathrm{m\,s^{-1}}$ wind, with a soarable band about 38 m deep (Fig. S1, supplementary). That exceeds the optimistic sink of a crude glider (0.75–$1.0\,\mathrm{m\,s^{-1}}$), so the slope can in principle hold the apparatus aloft, or at least flatten its descent, over the launch zone. The mechanism is present and of the right order; with the smoothed DEM, both the modelled slope and the lift it produces are lower bounds, which is why we keep the full field in the supplement and carry only its magnitude into the glide analysis.

4.4 A 200 m glide needs the wind

In still air, covering 200 m from an 18 m tower demands a glide ratio of 11.1, which is sailplane class and far beyond a wood-and-cloth wing (Fig. 4). Adding the modelled ridge lift collapses the requirement: above $3.6\,\mathrm{m\,s^{-1}}$ of southwest wind the required glide ratio drops into the crude cambered-wing band (L/D 3–7), reaching 4.8 at $8\,\mathrm{m\,s^{-1}}$. An L/D 5 wing launched into that wind clears 200 m at about $7.5\,\mathrm{m\,s^{-1}}$, inside the favourable band of Fig. 3. The distance is achievable, but only with the wind and slope doing part of the work; a calm-air leap from the tower would fall far short.

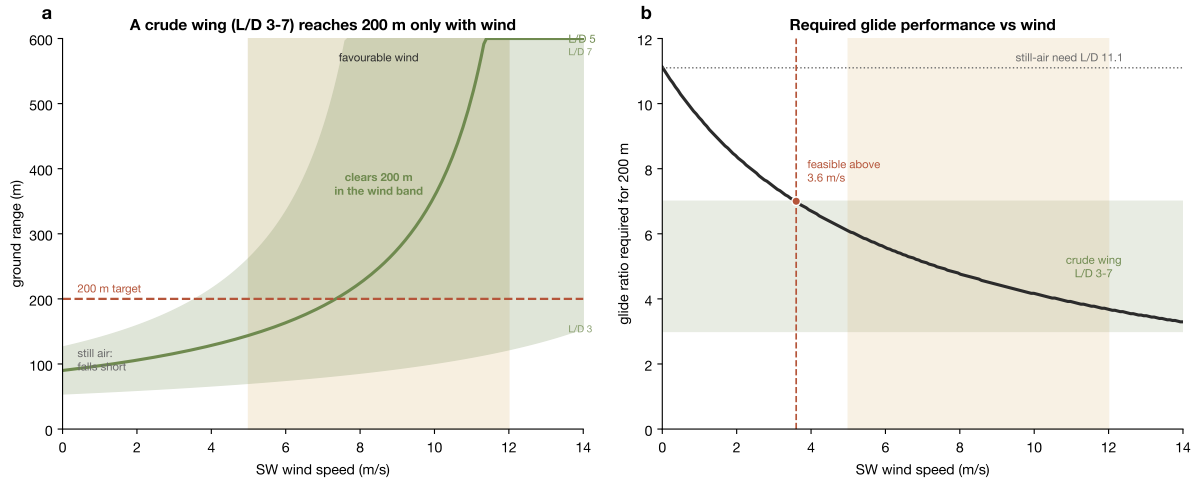


Figure 4: A 200 m glide needs wind and slope. (a) Range of an L/D 5 cambered wing launched from an 18 m tower versus southwest wind speed; 200 m is reached at about $7.5\,\mathrm{m\,s^{-1}}$. **(b)** Glide ratio required to cover 200 m versus wind speed. Still air demands L/D 11.1 (sailplane class, dotted line); with ridge lift the requirement falls into the crude cambered-wing band (L/D 3–7) above $3.6\,\mathrm{m\,s^{-1}}$, reaching L/D 4.8 at $8\,\mathrm{m\,s^{-1}}$.

4.5 The tailless craft cannot hold pitch

A hanging pilot's CG (0.35–0.55 c) sits aft of a tailless wing's neutral point (0.25 c), so the configuration is statically unstable for every posture the pilot can adopt (Fig. 5). The instability is fast: after an 8° gust the pitch attitude crosses the 60° loss-of-control threshold in 0.74 s, and the pitch error doubles in 0.1–0.5 s across the whole CG band, even with pitch inertia varied by a factor of two, against a representative flight lasting 13.3 s. Fitting a tail moves the neutral point aft to 0.53 c and adds strong pitch damping, which removes the divergence entirely. The aerodynamics single out exactly the part Eilmer said he had left off.

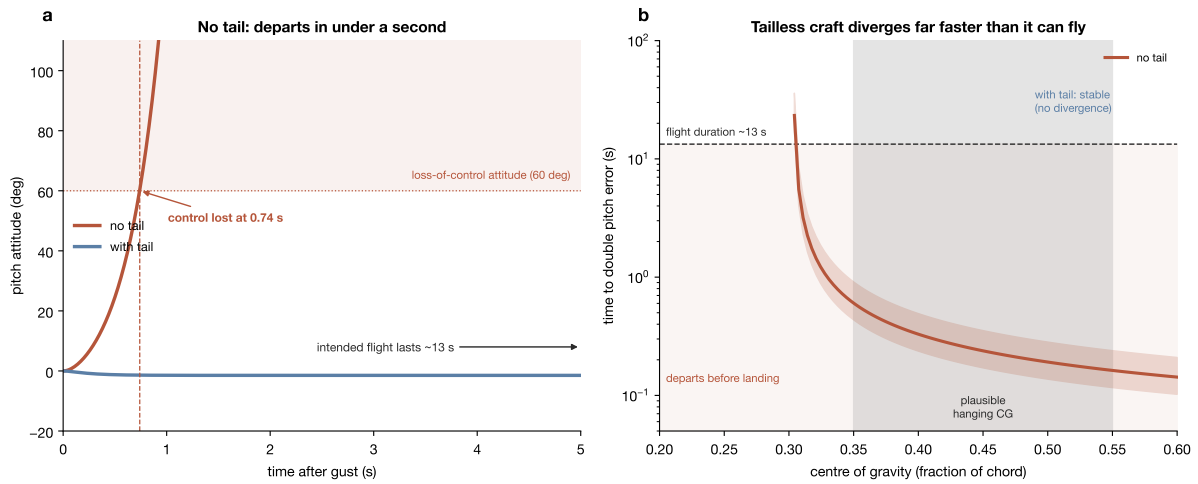


Figure 5: The tailless craft cannot hold pitch. (a) Pitch-attitude response to an 8° gust. Without a tail (red) the attitude crosses the 60° loss-of-control line 0.74 s after the gust; with a tail (blue) the same gust damps out. **(b)** Time to double the pitch error across the plausible CG band: 0.1–0.5 s everywhere for the tailless craft (pitch inertia varied 0.5–2×), far shorter than the flight, and no divergence with a tail. A tail shifts the neutral point from 0.25 c to 0.53 c, aft of a hanging pilot's CG (0.35–0.55 c).

4.6 The feasibility envelope

Sampling the unknown apparatus and launch day 40 000 times, 27 % of plausible worlds reach 200 m (95 % CI 26–27), with a median range of 147 m (Fig. 6a). The typical draw falls short, but a furlong is far from rare, and Eilmer chose his day and presumably his build, placing him in the upper tail. Whether a reaching flight survives turns entirely on the tail, through a clean timescale separation (Fig. 6b): a tailless craft's pitch error doubles in a median 0.27 s (p99 1.6 s), with control lost after about 3 m of flight, while the glide must stay controlled for a median 20 s, roughly 75× longer. Every tailed configuration that reaches 200 m is statically stable and never diverges. The model reproduces the reported sequence: he covered the distance, then lost control.

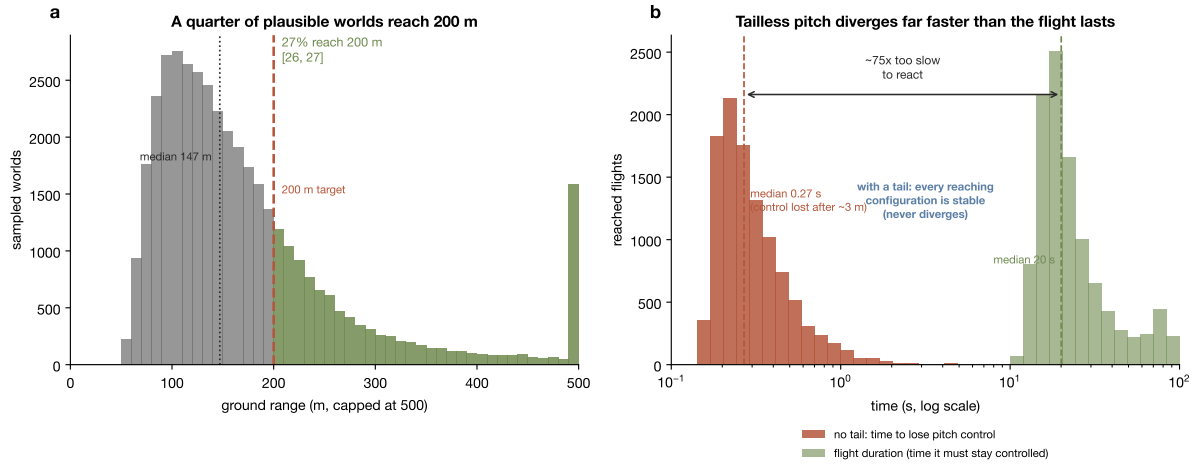


Figure 6: Feasibility envelope over the unknown apparatus and day. 40 000 Monte Carlo samples drawing apparatus and launch parameters from plausible ranges, with launch wind drawn from the actual ERA5 SW-sector launchable speeds. **(a)** 27 % of worlds reach 200 m (95 % CI 26–27; median range 147 m). **(b)** Among the reaching flights, tailless pitch error doubles in a median 0.27 s (p99 1.6 s) against a flight that must stay controlled for a median 20 s, while tailed configurations never diverge.

4.7 Broken legs fit an out-of-control arrival

The reported injury is itself evidence (Fig. 7). A flared touchdown arrives at the sink rate (1–3 $\mathrm{m\,s^{-1}}$), which is far below the tibia fracture-onset speed of about 9.7 $\mathrm{m\,s^{-1}}$ (range 7.5–11.5), so a controlled landing could not break the legs. An uncontrolled departure arrives near flying speed, with a vertical component of $V \sin\theta$, and breaks both tibiae with probability 0.66, yet at about 28 g over the legs it stays within the survivable feet-first regime rather than the fatal one. The injuries are thus positive evidence that Eilmer arrived out of control, the same failure mode the missing tail predicts in Figs. 5 and 6.

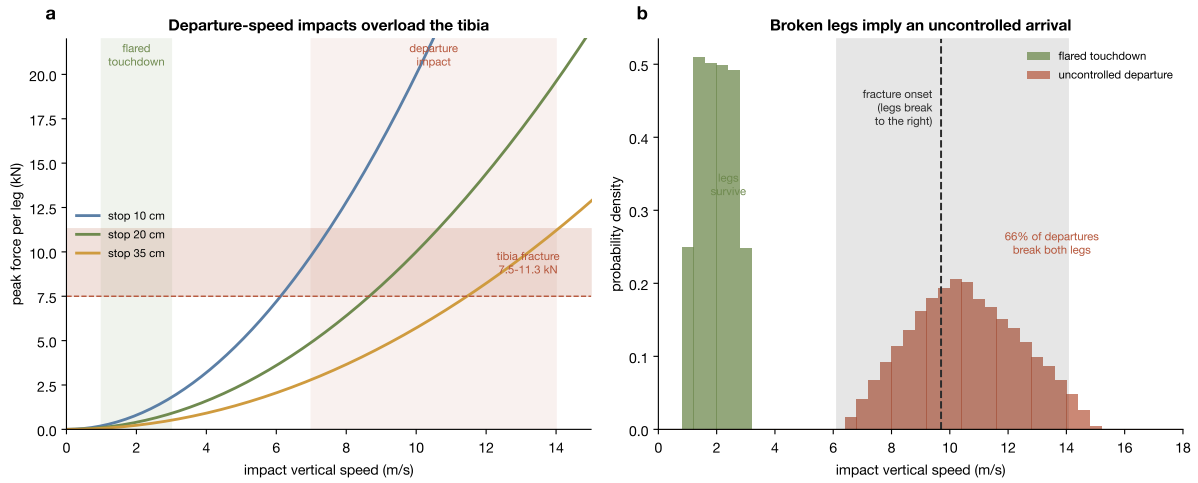


Figure 7: Broken legs as evidence of an out-of-control arrival. (a) Average feet-first impact force per leg versus vertical impact speed for ground-give distances $d = 10$–35 cm, against the axial tibia fracture band (7.5–11.3 kN per leg). Fracture onset is about 9.7 $\mathrm{m\,s^{-1}}$. **(b)** Impact-speed distributions for a flared touchdown (1–3 $\mathrm{m\,s^{-1}}$) versus an uncontrolled departure ($V \sin\theta$, $V = 11$–$16\ \mathrm{m\,s^{-1}}$, $\theta = 35°$–$70°$). A departure breaks both tibiae with probability 0.66 yet, at about 28 g, remains survivable.

5 Discussion

The account survives every test we can pose. The site geometry, the wind climatology, the glide physics, the pitch stability, and the impact biomechanics line up with what William reports, and they converge on the same cause of failure that Eilmer himself named. Some of these results hold regardless of our modelling choices and some depend on them, and the two should not be read with equal confidence.

What holds firm does not depend on the fine detail of the apparatus. The escarpment faces the prevailing wind and favourable launch conditions are common (Figs. 2 and 3); this is measured, not modelled. The static instability of a tailless hand-and-foot wing is similarly hard to escape: it follows from the pilot's CG lying aft of the wing's neutral point, which holds for every posture and every plausible wing in the Monte Carlo (Figs. 5 and 6). And the qualitative biomechanics are one-directional: a gentle landing cannot break two legs, so the injury implies speed at impact (Fig. 7).

What is model-dependent is the quantitative margin. The ridge-lift magnitude rests on a first-order linear model over a smoothed 6 m DEM (Eq. (1)), so the 1.5 $\mathrm{m\,s^{-1}}$ upwash and the 3.6 $\mathrm{m\,s^{-1}}$ wind threshold are order-of-magnitude statements that a LIDAR-resolved large-eddy solve should sharpen. The 27 % reach fraction is a function of the priors we placed on wing area, mass, and glide ratio, and should be read as the spread of a plausible space rather than

a calibrated probability. The stability analysis is a linear short-period treatment of the pitch axis alone; the lateral modes and the nonlinear post-departure motion need a full six-degree-of freedom model before the 75× timescale separation can be called exact.

The Ibn Firnas parallel deserves a closer look in light of these results. If two of the earliest credible glide attempts both ended in a tailless loss-of-control arrival [11], the convergence is not coincidence but physics: a tail is the one component an untutored builder is most likely to omit and the one whose absence most reliably ends the flight. The method here, terrain plus climate plus first-principles flight dynamics, can be pointed at that account and others on the same footing.

The main threats to the conclusion are the unknown apparatus and the coarse terrain. We mitigate the first with the Monte Carlo and the second by reporting slope and lift as lower bounds, but neither is a substitute for the two obvious next steps: an airborne-LIDAR terrain model with a microscale wind solve, and a structural analysis of a willow-and-cloth wing under the computed flight loads to rule out a structural, rather than control, failure. We also do not reconstruct the actual flight. We show that the reported flight is physically consistent and that its stated cause is correct, which is a different and more defensible claim than asserting it happened exactly as written.

6 Conclusion

William of Malmesbury's account of Eilmer's flight is aerodynamically self-consistent. A wind-and-slope-assisted glide of about 200 m from the abbey is feasible for a crude cambered wing, the tailless configuration Eilmer blamed would by itself cause loss of pitch control within a second of a disturbance, and the broken legs he suffered fit the predicted out-of-control impact while remaining survivable. The monk's own diagnosis, that he should have fitted a tail, is correct.

We do not claim the flight happened exactly as written, and the quantitative margins depend on a coarse terrain model and a first-order lift model that airborne LIDAR and a microscale wind solve would sharpen. What we provide is a calibrated verdict on a specific eleventh-century claim and a reproducible pipeline, from terrain and climate to flight dynamics and biomechanics, for adjudicating pre-modern flight accounts. The same method applied to Ibn Firnas and the other early gliding reports would test whether the want of a tail is the recurring cause it appears to be.

Data and code availability

The full analysis pipeline, raw data manifests, and figure-generation scripts rebuild every number and figure in this paper from one driver script, and are released with the manuscript.

Acknowledgements

We thank the Athelstan Museum, Malmesbury, for curating the local record of Eilmer's flight.

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Supplementary material

S1. Verbatim source extracts for load-bearing claims

Three claims carry the argument: the primary account of the flight, the tibia fracture threshold behind the biomechanics, and the medieval wind regime behind the climatology. We quote each source directly so the reader can check our reading.

S1.1 The flight, the distance, the injury, and the tail. The entire factual basis is one passage of William of Malmesbury, *Gesta Regum Anglorum* (book II, ch. 225), in the Oxford Medieval Texts edition [1]. The Latin reads:

> *Is erat litteris, quantum ad id temporis, bene imbutus, aeuo maturus, immanem audaciam prima iuuentute conatus: nam pennas manibus et pedibus haud scio qua innexuerat arte, ut Daedali more uolaret, fabulam pro uero amplexus, collectaque e summo turris aura, spatio stadii et plus uolauit; sed uenti et turbinis uiolentia, simul et temerarii facti conscientia, tremulus cecidit, perpetuo post haec debilis, et crura effractus. Ipse ferebat causam ruinae quod caudam in posteriori parte oblitus fuerit.*

In the standard English rendering:

> He had by some means, I scarcely know what, fastened wings to his hands and feet so that, mistaking fable for truth, he might fly like Daedalus, and, collecting the breeze on the summit of a tower, he flew for more than the distance of a furlong. But, agitated by the violence of the wind and the swirling of air, as well as by awareness of his rashness, he fell, broke his legs, and was lame ever after. He himself used to say that the cause of his failure was his forgetting to put a tail on the back part.

Every element we model is in these two sentences: the tower launch, the hand-and-foot wings, the breeze, the furlong (a stadium, about 200 m), the loss of control in gusty air, the broken legs and permanent lameness, and the diagnosis (*caudam . . . oblitus*, "forgetting a tail").

S1.2 The tibia fracture threshold. The 7.5–11.3 kN per-leg band in Fig. 7 comes from the injury-biomechanics literature, consolidated in a recent systematic review [24], which reports that

> tibial fracture thresholds range from ∼7.5 kN in female specimens to 11.3 kN under combined axial and bending loads, with fibula contribution increasing axial tolerance by ∼10%.

The upper figure traces to the primary compilation of Salzar, Lievers, and Bailey [23], which recommends a tibial axial threshold of ∼10.3 kN rising to ∼11.3 kN with the fibula's contribution. We use the range, not a point value, precisely because the threshold depends on sex, loading rate, and bending.

S1.3 The medieval wind regime. The claim that modern reanalysis is a reasonable, probably conservative proxy for Eilmer's epoch rests on reconstructions of the North Atlantic Oscillation, which disagree on timing. Trouet et al. [16] report

> a 947-year-long multidecadal North Atlantic Oscillation (NAO) reconstruction [with] a persistent positive NAO during the MCA [Medieval Climate Anomaly].

The later, model-tested reconstruction of Ortega et al. [17] revises this, finding

> no persistent positive NAO during the medieval period, but [suggesting] that positive phases were dominant during the thirteenth and fourteenth centuries.

We therefore make only the weaker claim the evidence supports: neither reconstruction indicates the weak westerlies of a negative NAO over Britain in Eilmer's period, so the favourable-wind fraction we compute from 2004–2023 ERA5 (Fig. 3) is not an overestimate for the eleventh century. We do not rely on medieval winds being *stronger* than modern ones.

S2. Monte Carlo parameter priors

Table S1 lists the plausible ranges sampled in the feasibility envelope (Fig. 6). All priors are uniform over the stated range.

Table S1: Sampled parameter ranges for the Monte Carlo (Fig. 6). Launch wind is not sampled from a prior but drawn from the empirical ERA5 SW-sector launchable speeds (Fig. 3).

Parameter	Range	Basis
Wing area	5–10 m^2	crude hand/foot wing
All-up mass	70–90 kg	adult pilot plus apparatus
Trim C_L	0.7–1.1	cambered cloth, low Reynolds number
Glide ratio L/D	3–7	hang-glider class, crude build
Tower height	15–25 m	abbey tower, conservative
Escarpment slope	8°–18°	DEM p90 as a floor
Ridge-lift efficiency	0.5–1.0	attached-upslope fraction
Centre of gravity	0.35–0.55 c	hanging-pilot range
Aspect ratio	3–5	span/area of a hand-foot wing
Tail neutral-point shift	0.18–0.32 c	tail size variation

S3. Ridge-lift field

Figure S1 shows the first-order orographic lift field summarised in the main text. We keep the full field here because the smoothed 6 m DEM makes the modelled slope, and therefore the lift, a lower bound; the main analysis carries only its magnitude.

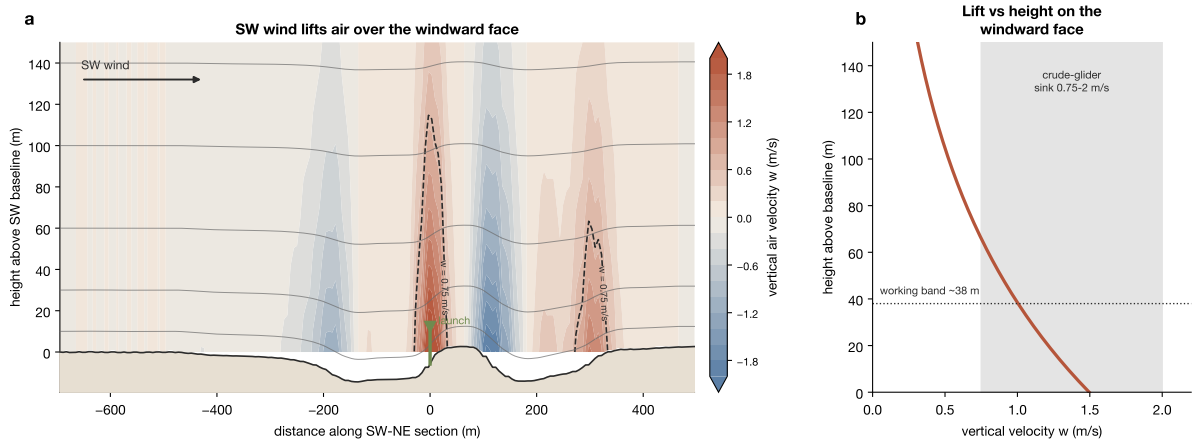


Figure S1: Ridge lift over the escarpment matches a crude glider's sink. A first-order linear orographic model ($w = U\,\mathrm{d}h/\mathrm{d}x\,e^{-kz}$) over the real SW–NE terrain section. **(a)** Vertical-velocity field and streamlines for an $8\,\mathrm{m\,s^{-1}}$ southwest wind. **(b)** Lift versus height above the windward face. Peak lift just southwest of the abbey is about $1.5\,\mathrm{m\,s^{-1}}$, above a crude glider's optimistic sink of 0.75–$1.0\,\mathrm{m\,s^{-1}}$, with a soarable band about 38 m deep.